THE CHAIN ATOM $W = u^5v + v^6$: MILNOR NUMBER 25, DIAGONAL SYMMETRY 30, AND THE TRANSPOSE SHIFT $+1$

BENJAMIN STANLEY FROHMAN

ABSTRACT. Let $W(u, v) = u^5v + v^6$. This is an isolated invertible plane-curve singularity of chain type. Two integers that sit next to each other in the exponent matrix were conflated in earlier notes: the order of the maximal diagonal symmetry group is $|\det A| = 30$, while the Milnor number—the dimension of the Jacobian algebra—is $\mu(W) = 25$. The Berglund–Hübsch transpose $W^T = u^5 + uv^6$ has the same symmetry order $|\mathrm{Aut}(W^T)| = 30$ and the same central charge $\hat{c} = 4/3$, but Milnor number $\mu(W^T) = 26 = \mu(W) + 1$. Three disjoint Thom–Sebastiani copies of W reconstruct the sextic

$$F = x_0^5x_3 + x_3^6 + x_1^5x_4 + x_4^6 + x_2^5x_5 + x_5^6,$$

with $\mu(F) = 25^3 = 15625$ and $|\mathrm{Aut}(F)| = 30^3 = 27000$. Steenbrink spectra and monodromy characteristic polynomials are computed from the standard monomials of the Jacobian ideals. This note records those computations; it does not claim a rational Hodge miss on the fourfold $V(F)$.

1. The Germ and Diagonal Symmetries

Consider the invertible polynomial of chain type

$$W(u, v) = u^5v + v^6 = v(u^5 + v^5), \qquad A = \begin{pmatrix} 5 & 1 \\ 0 & 6 \end{pmatrix}, \qquad |\det A| = 30.$$

Diagonal symmetries $(\lambda, \mu) \in (\mathbb{C}^*)^2$ satisfying $W(\lambda u, \mu v) = W(u, v)$ obey $\mu^6 = 1$ and $\lambda^5\mu = 1$, yielding $|\mathrm{Aut}(W)| = |\det A| = 30$ characters. The quasihomogeneous weights are $(1, 1)$ of degree 6, charges $q = (1/6, 1/6)$, and central charge $\hat{c} = \sum(1 - 2q_i) = 4/3$. The partial derivatives

$$\frac{\partial W}{\partial u} = 5u^4v, \qquad \frac{\partial W}{\partial v} = u^5 + 6v^5$$

vanish simultaneously only at $(0, 0)$, confirming that W is isolated at the origin.

2. Jacobian Algebra and Milnor Number $\mu(W) = 25$

In characteristic 0, $v^6 \in (\partial W)$ since $v(\partial_v W) - \frac{1}{5}u(\partial_u W) = 6v^6$. A reduced Gröbner basis over $\mathbb{Q}$ is

$$\big(v^6,\ u^5 + 6v^5,\ u^4v\big),$$

with leading monomials v^6, u^5, and u^4v. The standard monomials of the Jacobian algebra $J_W = \mathbb{C}[u, v]/(\partial_u W, \partial_v W)$ are the 25 monomials

$$\{u^iv^j : 0 \le i \le 3,\ 0 \le j \le 5\} \ \cup\ \{u^4\}.$$

Date: September 25, 2026.

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This matches the Milnor–Orlik quasihomogeneous formula $\mu(W) = (6/1 - 1)^2 = 25$. In particular, $\mu(W) \neq |\mathrm{Aut}(W)|$: the integer 30 is the diagonal group order, whereas 25 is the Jacobian dimension. On the chain $u^5v + v^q$ ($q = 6$), the extra leading term u^4v removes $u^4v, \ldots, u^4v^5$ from the 5×6 box, giving $\mu = |\det A|(1 - 1/q) = 30 \cdot \frac{5}{6} = 25$.

3. The Berglund–Hübsch Transpose W^T and the +1 Shift

The Berglund–Hübsch transpose has exponent matrix A^T:

$$W^T(u, v) = u^5 + uv^6, \qquad A^T = \begin{pmatrix} 5 & 0 \\ 1 & 6 \end{pmatrix}.$$

It is quasihomogeneous with weights $(3, 2)$ and degree 15, preserving $\hat{c} = 4/3$ and $|\mathrm{Aut}(W^T)| = 30$. Its partials are $\partial_u W^T = 5u^4 + v^6$ and $\partial_v W^T = 6uv^5$. The eliminant identity

$$v^5(5u^4 + v^6) - 5u^3(uv^5) = v^{11}$$

shows that the isolated pure-power and mixed leading monomials of (∂W^T) are

$$u^4, \qquad uv^5, \qquad v^{11}.$$

The standard monomials are therefore the $11 + 3 \cdot 5 = 26$ monomials

$$\{v^0, v^1, \ldots, v^{10}\} \cup \{u, u^2, u^3\} \times \{v^0, v^1, v^2, v^3, v^4\},$$

matching the weighted formula $\mu(W^T) = (15/3 - 1)(15/2 - 1) = 4 \cdot \frac{13}{2} = 26$. Hence

$$\mu(W^T) = \mu(W) + 1.$$

Quantity	$\lvert\det A\rvert$	μ
Meaning	Diagonal symmetry order	Jacobian dimension / vanishing cycles
Chain atom $W = u^5v + v^6$	30	25
Transpose $W^T = u^5 + uv^6$	30	26
Three-block sum F	$30^3 = 27000$	$25^3 = 15625$
Preserved by BHK transpose?	Yes	No (+1 shift)

4. Steenbrink Spectrum and Monodromy

For W (weights $(1, 1)$, degree 6), the Steenbrink spectral numbers $\alpha = (i + j + 2)/6$ on the 25 standard monomials are

$$\tfrac{1}{3},\ \tfrac{1}{2},\ \tfrac{2}{3},\ \tfrac{5}{6},\ 1,\ \tfrac{7}{6},\ \tfrac{4}{3},\ \tfrac{3}{2},\ \tfrac{5}{3}$$

with multiplicities $1, 2, 3, 4, 5, 4, 3, 2, 1$ (sum 25, mean 1, symmetric about 1). The characteristic polynomial of monodromy on H_1 of the Milnor fiber is

$$\Delta_W(t) = (t - 1)^5(t + 1)^4(t^2 + t + 1)^4(t^2 - t + 1)^4 = (t^6 - 1)^4(t - 1).$$

For W^T (weights $(3, 2)$, degree 15), the 26 spectral numbers lie in $[1/3, 5/3]$ with mean 1 and an extra $+1$ at the center, yielding

$$\Delta_{W^T}(t) = \Phi_1(t)^2\Phi_3(t)^2\Phi_5(t)\Phi_{15}(t)^2 = \frac{(t^{15} - 1)^2(t - 1)}{t^5 - 1}.$$

5. Why Incomplete Leading-Term Lists in a Box Print 31 and 35

In degree-graded reverse lexicographic order (`grlex`), the generator $5u^4 + v^6$ has leading term v^6 in unweighted degree, or $5u^4$ if v^{11} is not extracted as a separate leading power. Reducing a finite box $\{0 \le i, j < N\}$ only against the incomplete pair $\{u^4, uv^5\}$ leaves the v-axis unbounded and counts $N + 3 \cdot 5 = N + 15$ monomials: for $N = 16$ this prints 31, and for $N = 20$ it prints 35. Enforcing $v^{11} \in (\partial W^T)$ replaces N by 11 and recovers the true invariant $\mu(W^T) = 11 + 15 = 26$.

6. Scope Note

This note is a local singularity and Jacobian-algebra classification for (W, W^T). It does not construct a Fourier–Mukai partner fourfold $Y \not\cong V(F)$, does not construct a non-algebraic rational $(2, 2)$ Hodge class on $V(F)$, and does not claim a proof or disproof of the rational Hodge conjecture.

References

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Independent Researcher, ORCID: 0009-0006-7068-3718
Email address: `frohmanbenjamin@gmail.com`